Al-Balqa Applied University



جامعة البلقاء التطبيقية

17 x 1

**Problem # 1:** Multiple choice question: **[10 x 1pt]**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| B (5) | B | D | A | D | B | A | A | A | A |

1. Find the content of AL and ZF register after execution of the following code:

```
MOV AL,FCH
TEST AL,16
```

A. AL=FCH; ZF=1 **B.** AL=FCH; ZF=0 C. AL=00H; ZF=1 D. AL=0CH; ZF=0

1111 1100
0001 0000
0000 0000
ZF = 1

2. Instruction pointer (IP) contains the offset address of _____ segment.

A. Data segment **B.** Code segment C. Stack segment D. Extra segment

3. The output of the following operation on register BL is:

```
MOV BL,FCH
MOV CL,08
SHL BL,CL
```

**A.** BL= CFH B. BL= FCH C. BL= 0FH D. BL=00

1111 1100 << 8
0000 1000,
1100 0000

4. In the following instruction sequence, the values of the parity, auxiliary carry, and sign flags are:

```
MOV AX, 0C17FH
ADD AX, 3
```

A. PF=0, AF=1, SF=0 B. PF=1, AF=1, SF=1 C. PF=0, AF=0, SF=1 **D.** PF=1, AF=0, SF=0

PF=1, AF=0, SF=0

0C17F
+ 00003
0C182

5. If the offset address of X in data segment is equal to 230H, and DS=2500H, then the physical address of X is?

A. 20315 H **B.** 25320H C. 2500 H D. 323000

25000 + 320
= 25320

6. Which of the following is an example of a relative based addressing mode?

**A.** MOV AX, [BP + SI + 10h] B. MOV AX, [BP+1] C. MOV AX, [SI+1] D. MOV AX, [BX]

7. Suppose X defined as below; what is the content of CX, and AH registers after execution of the following instructions:

```
X    DW 4580H,0ABCH,1100,2222H
     MOV BX, OFFSET X
     MOV AH,1
     MOV CX,[BX]
     INC BX
     ADD AH, [BX]
```

| X | BX |
|---|---|
| +0 | 4580 |
| +2 | 0ABC |
| +4 | 1100 |
| +6 | 2222 |

CX = 4580H

AH = 01 + 0ABCH
= 0ABDH

**A.** CX=4580H; AH=46H B. CX=0ABCH; AH=0AH
C. CX=BCD0H; AH=B0H D. CX=BC45H; AH=0CH

8. Assuming DS=4000h, SI=10h, and BX=23h, the physical address of the source operand in the instruction MOV AX, [BX + SI + 0C3h] is,

**A.** 400F6h B. 4FFF6h C. 40033h D. 40123h

10 + 23 + C3 = F6
40000 + F6 = 400F6

9. The value of AX at the end of execution of the following 8086 assembly instruction sequence is:

```
MOV AL, -9
SHL AX, 4
```

**A.** AL=70h B. AL=0Fh C. AL= 60h D. AL= 80h

-9 = F7 (in AL)
AX = 00F7
SHL 4 → 0F70

10. In the following instruction sequence, show the value of AL:

```
MOV AL,0FFH
NEG AL
XOR AL,FFH
```

**A.** AL= FEH B. AL= F15H C. AL= 0FH D. AL= 0FFH

AL = FF
NEG → 01
XOR FF → FE

**Problem # 2:** Find the contents of the registers written below for each of the following assembly codes: **[10 pts]**

**A)**

```
MOV AX,0FFE0H
MOV CX,0000
PUSH AX
PUSH BX
POP DX
MOV X,DX
POP CX
HLT
```

| 0010 |
|---|
| 000E |
| ?? |
| FF |
| |

AX = FFE0H  CX = FFE0  DX = 0000

BX = 0000 H  SI = 0000 H

**B)**

```
X DB 1,2,3,4,5,6,7,9
MOV CX,8
MOV SI,0
L2: CMP X[SI],6
JB L
ADD AL,X[SI]
L: INC SI C
CMP SI,4
```

AL = 06 H  SI = 00 H

**C)**

```
X DW 35FAH, 48CBH, 231AH, 0080H
    MOV BX, OFFSET X
    ADD BX,2
    MOV AL,[BX]
    INC BX
    MOV DX,[BX]
```

| X | FA | 35 |
|---|---|---|
| | 31 | 37 |
| | CB | 39 |
| | 48 | 3B |
| | 1A | 3D |
| | 25 | 3F |
| | 80 | 41 |

AL = CB H  Dx = 1A48

BX = 0014H

**D)**

```
MOV AL,FEH
ADD AL,2
ADC AL,3
```

```
  1111 1110
+ 0000 0010
-----------
1 0000 0000
```

```
  CF = 1
+ 0000 0011
-----------
  0000 0100
```

AL = 04 H  ZF = 1

**Problem # 3:**

A) Find the result of DX without using multiplication and division instructions DX= 5*AX+9*BX

```
MOV CX, AX
MOV DX, BX
SHL AX, 2
SHL AX, 2
ADD AX, AX
MOV CL, 3
SHL BX, CL
```

```
MOV DX, AX
ADD DX, BX
```

DX = 5*AX + 9BX

DX = 4AX + 1AX

+ 8BX + BX

= 5AX + 9BX

B) Write the equivalent of the following instructions

- SUB AX,BX →

```
MOV DX, AX
NEG BX
ADD AX, BX
NEG BX
```

1

- XCHG AX,BX →

```
MOV DX, AX
MOV AX, BX
MOV BX, DX
```

2

- NOT AX →

```
XOR AX, FFFF
```

3